



Filter Wizard

Filter Wizard Design

Created on 06/18/2025



Filter Wizard Design Report

Filter Requirements for Low-Pass, 4th order Butterworth

Specifications: Optimize: Specific Parts; +Vs: 5; -Vs: -5

Gain: 0 dB

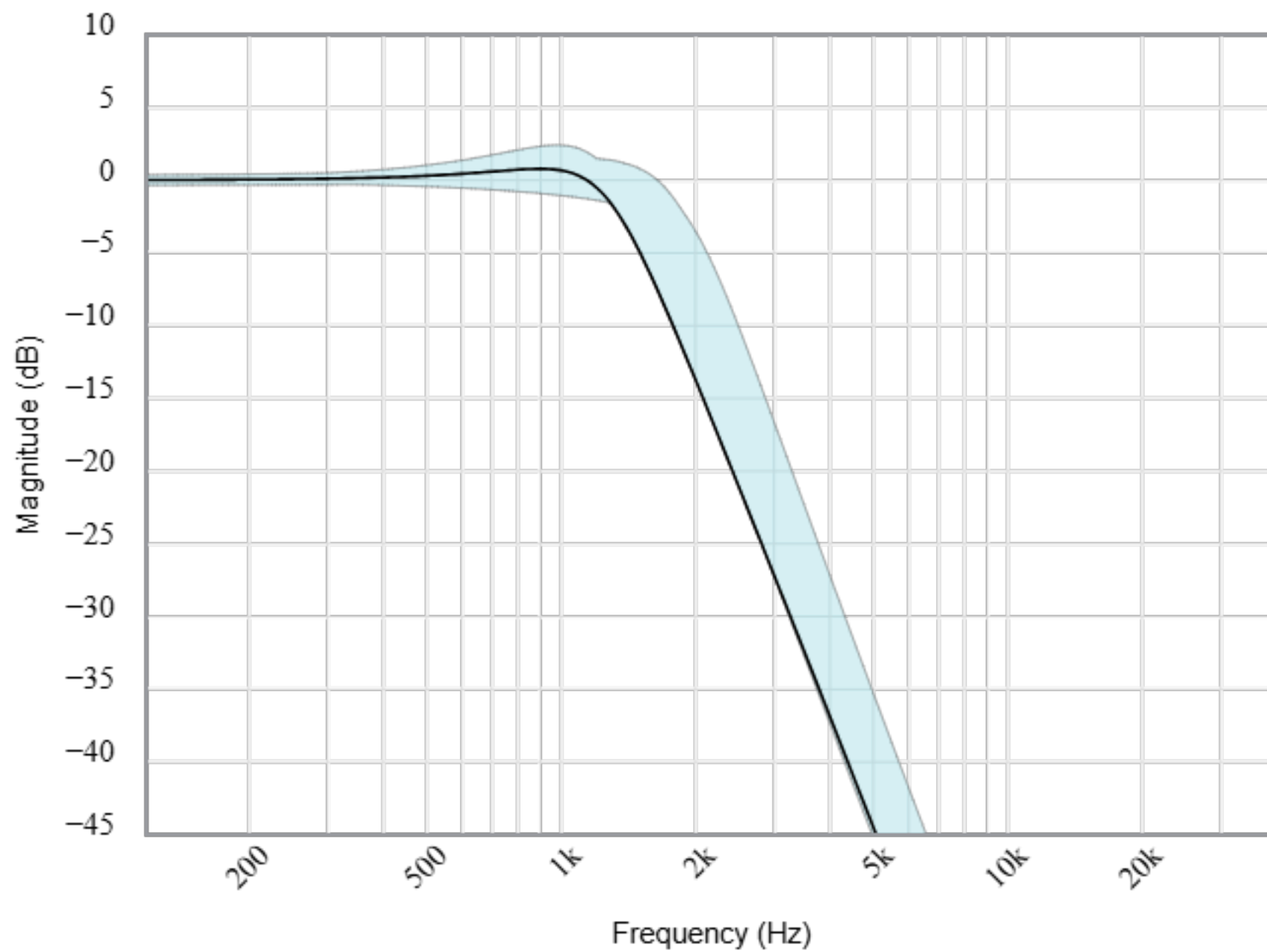
Passband: -0.5dB at 1.2kHz

Stopband: -25dB at 4kHz

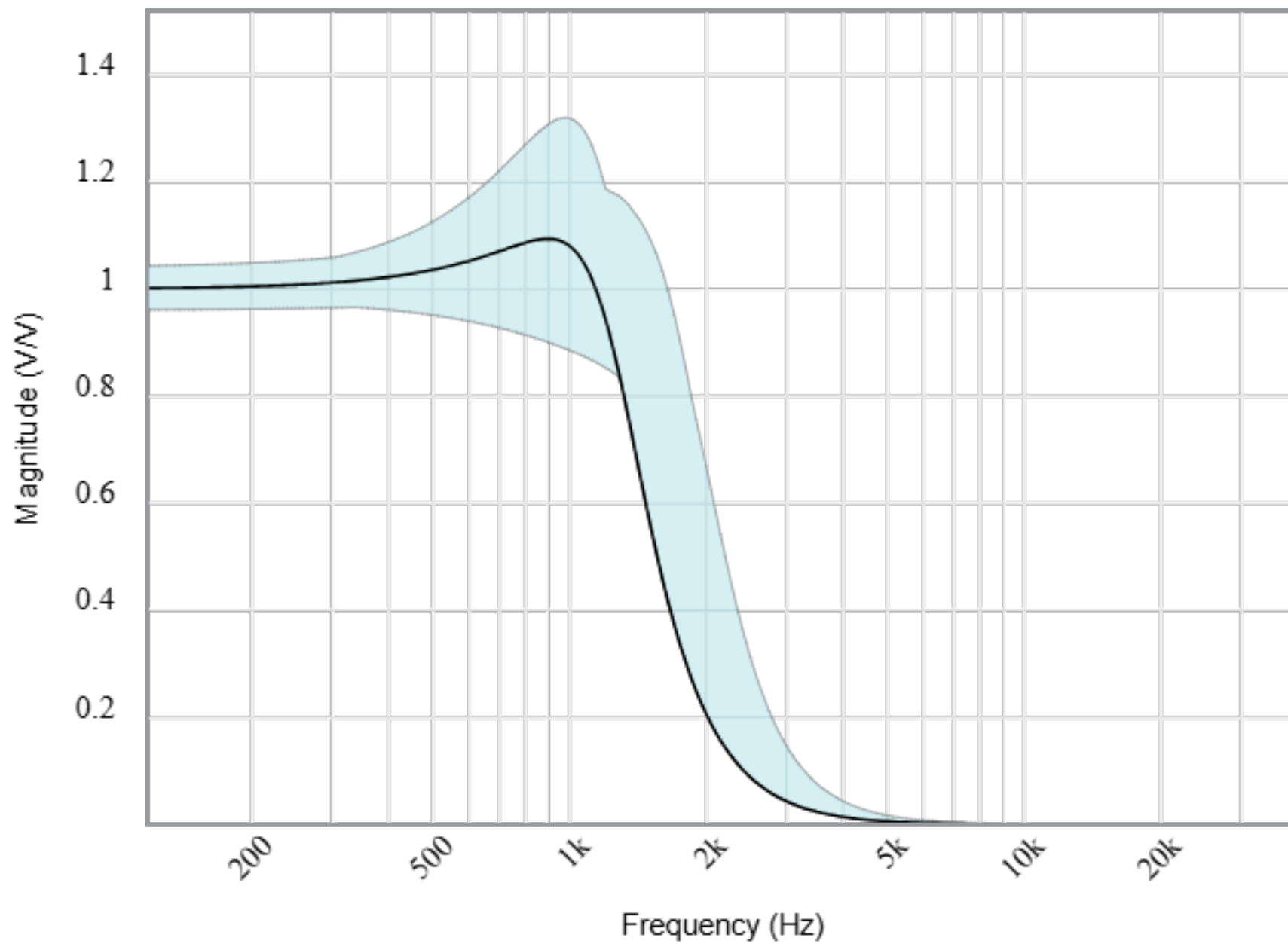
Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

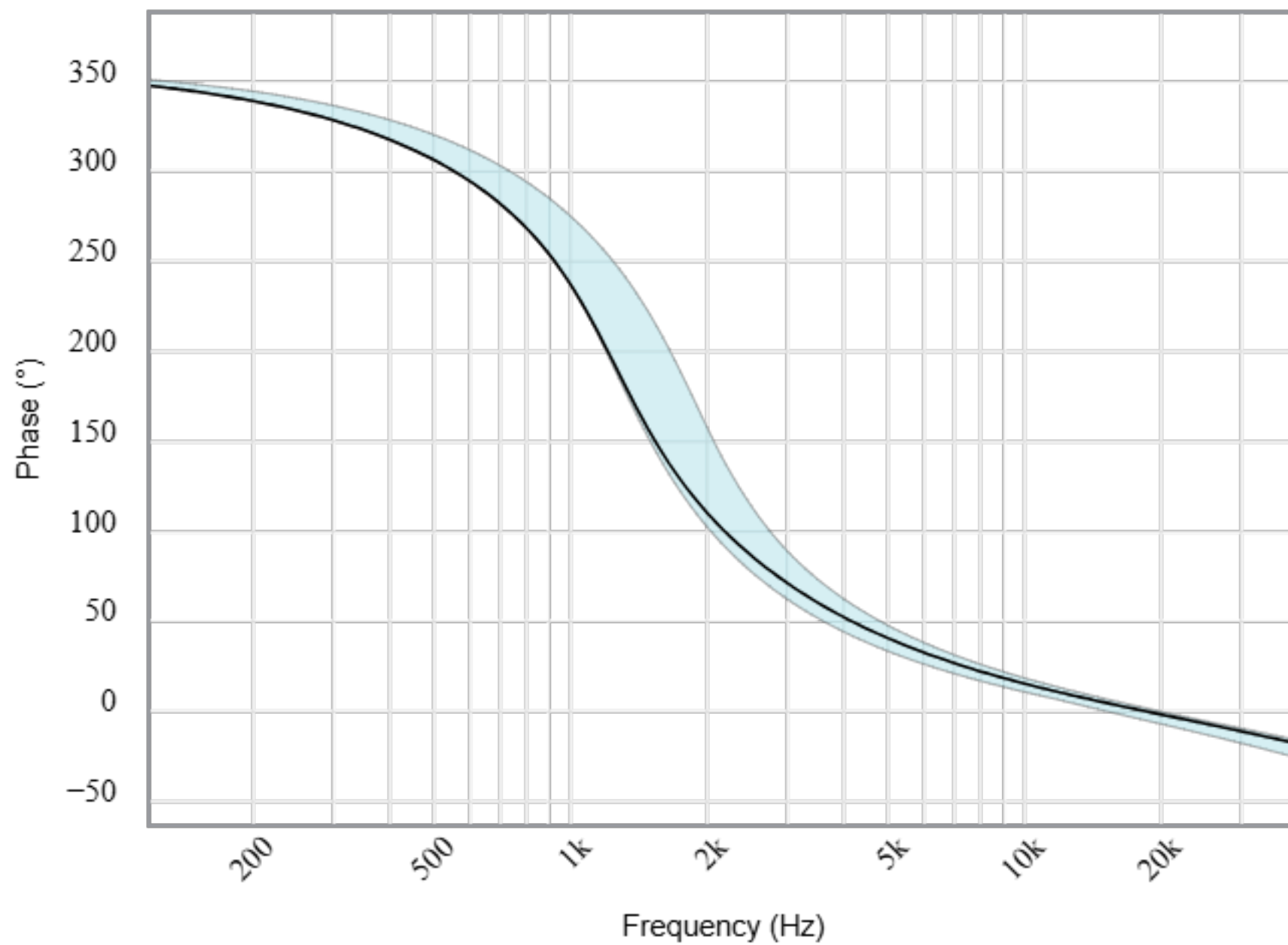
Magnitude(dB)



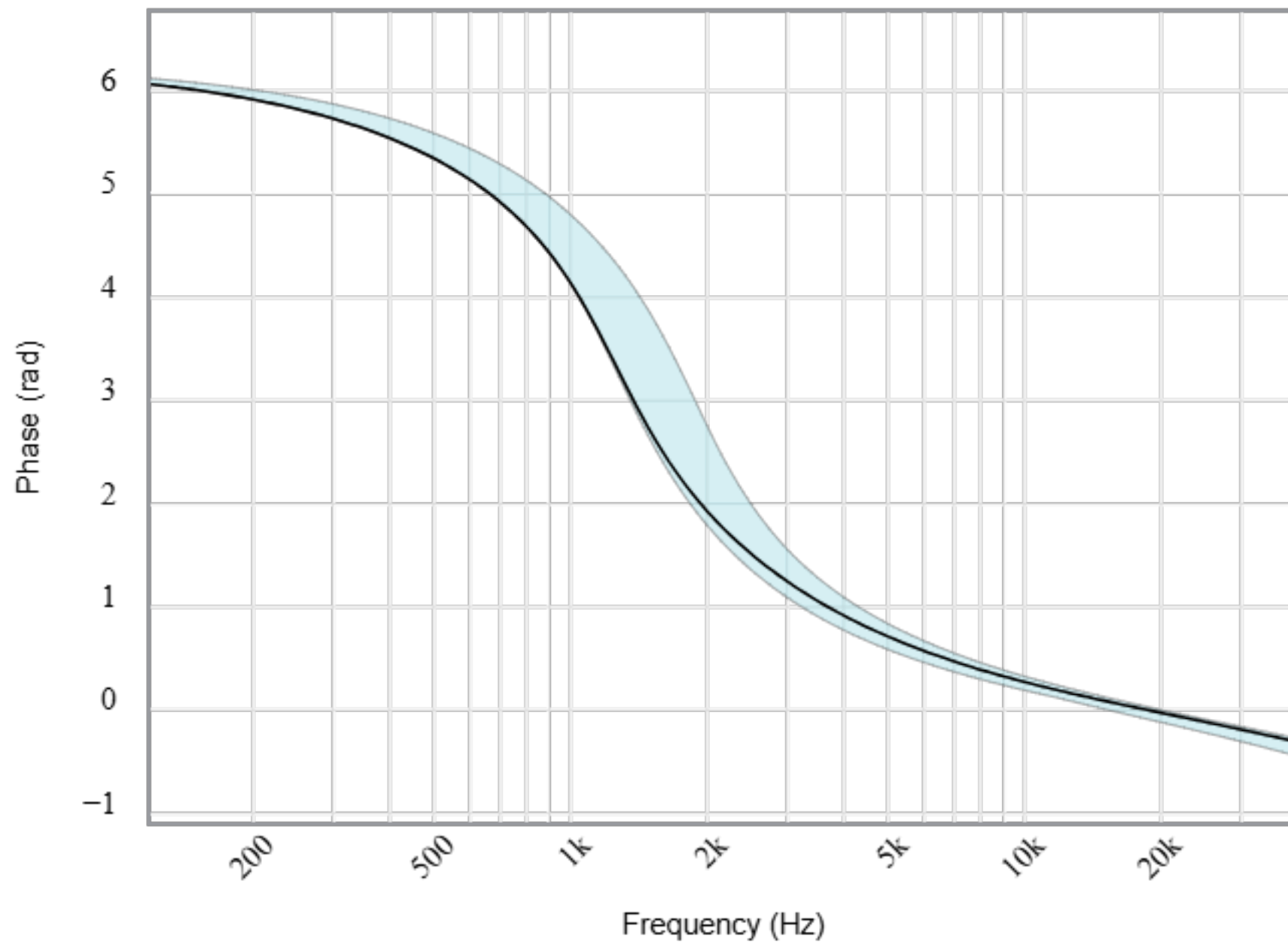
Magnitude(Volts per Volt)



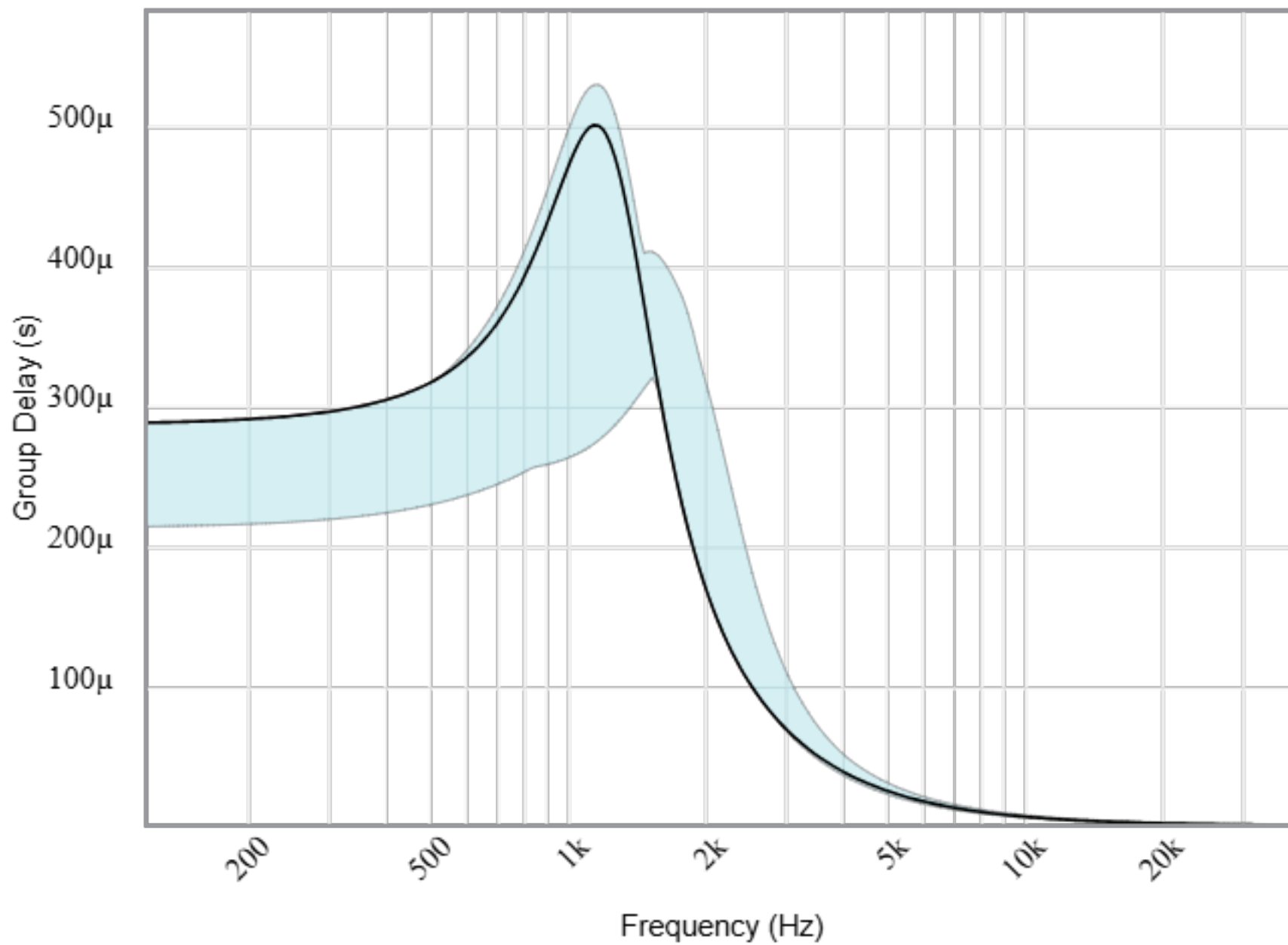
Phase(degrees)



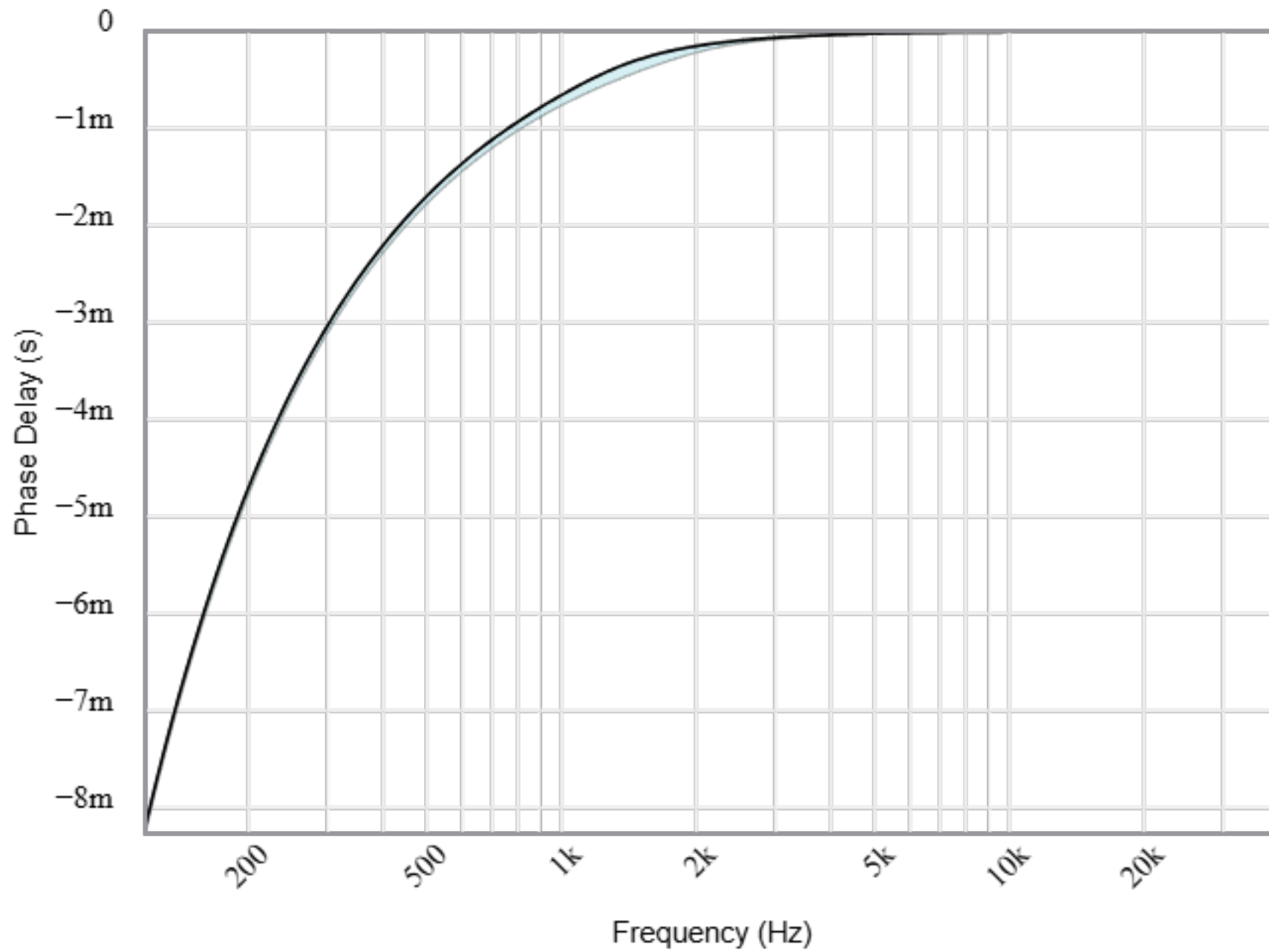
Phase(radians)



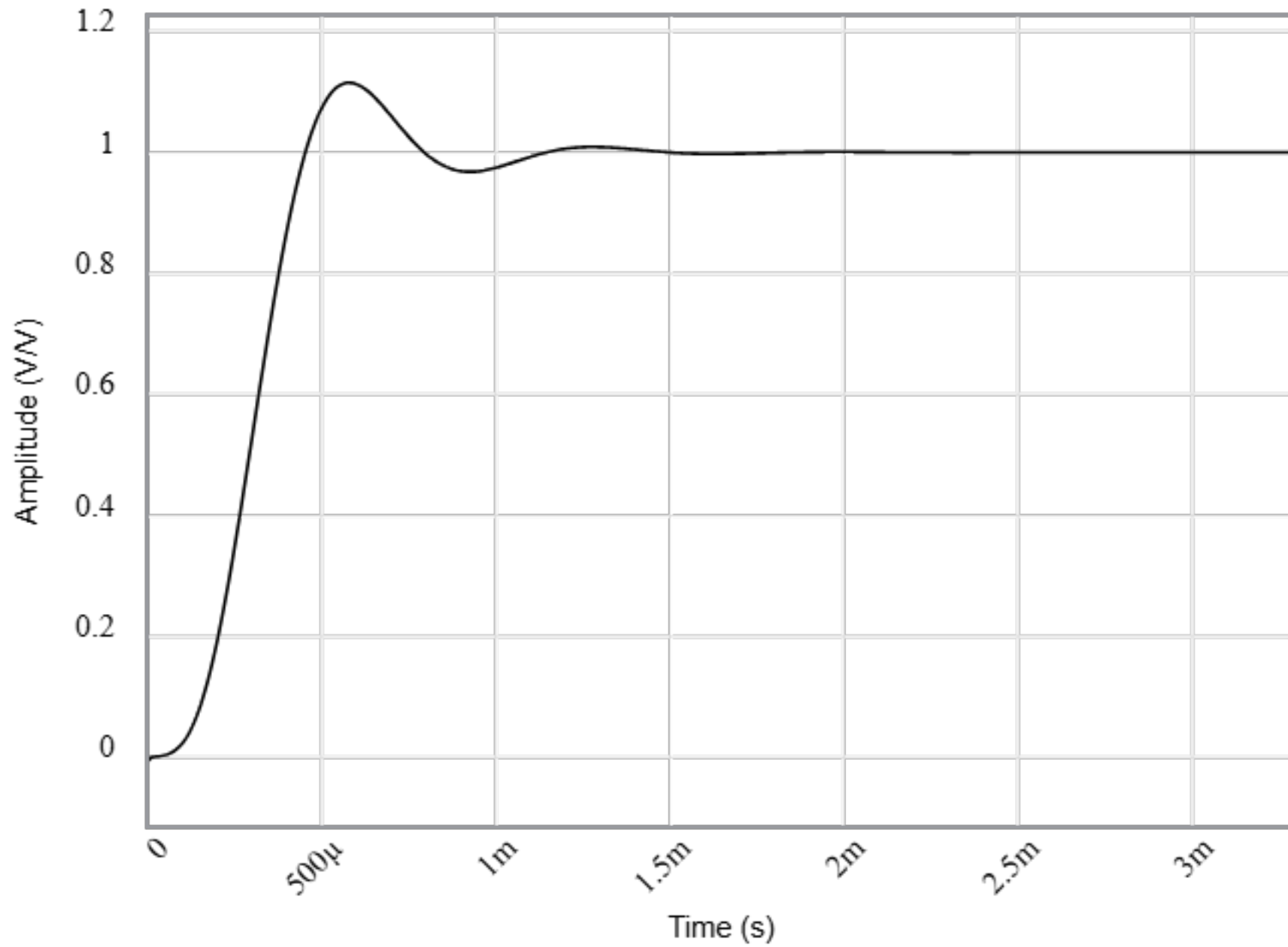
Group Delay



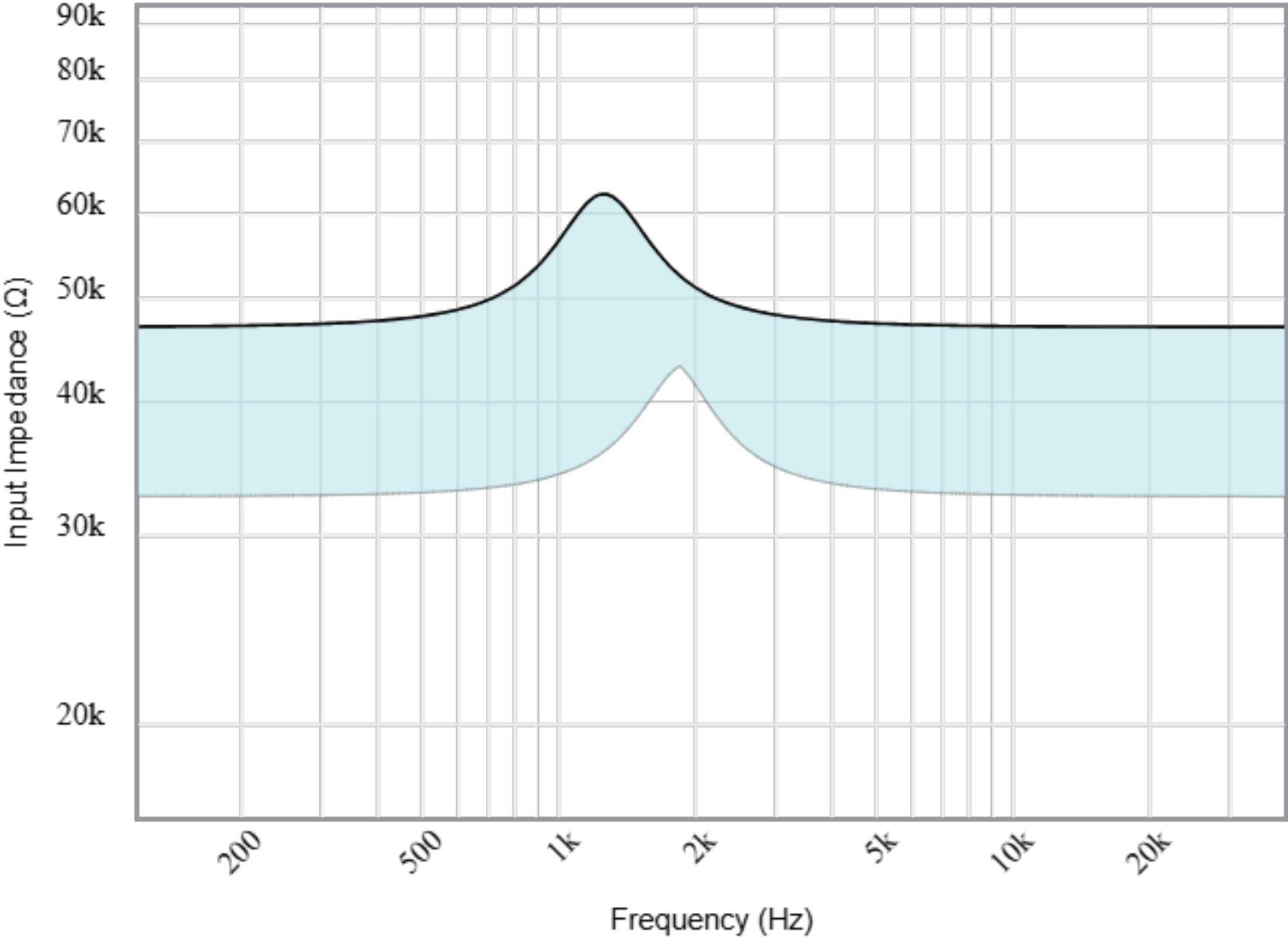
Phase Delay



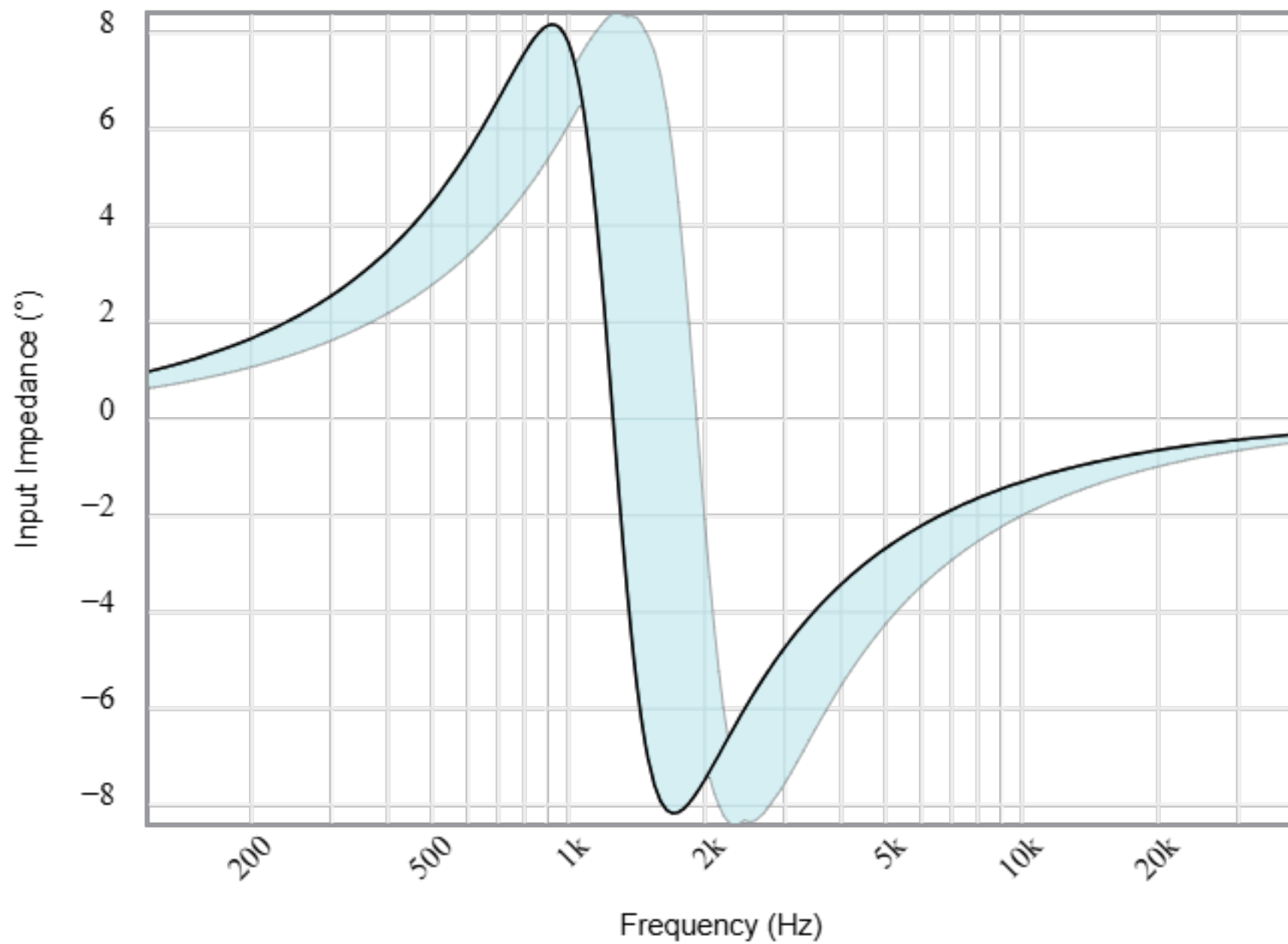
Step Response



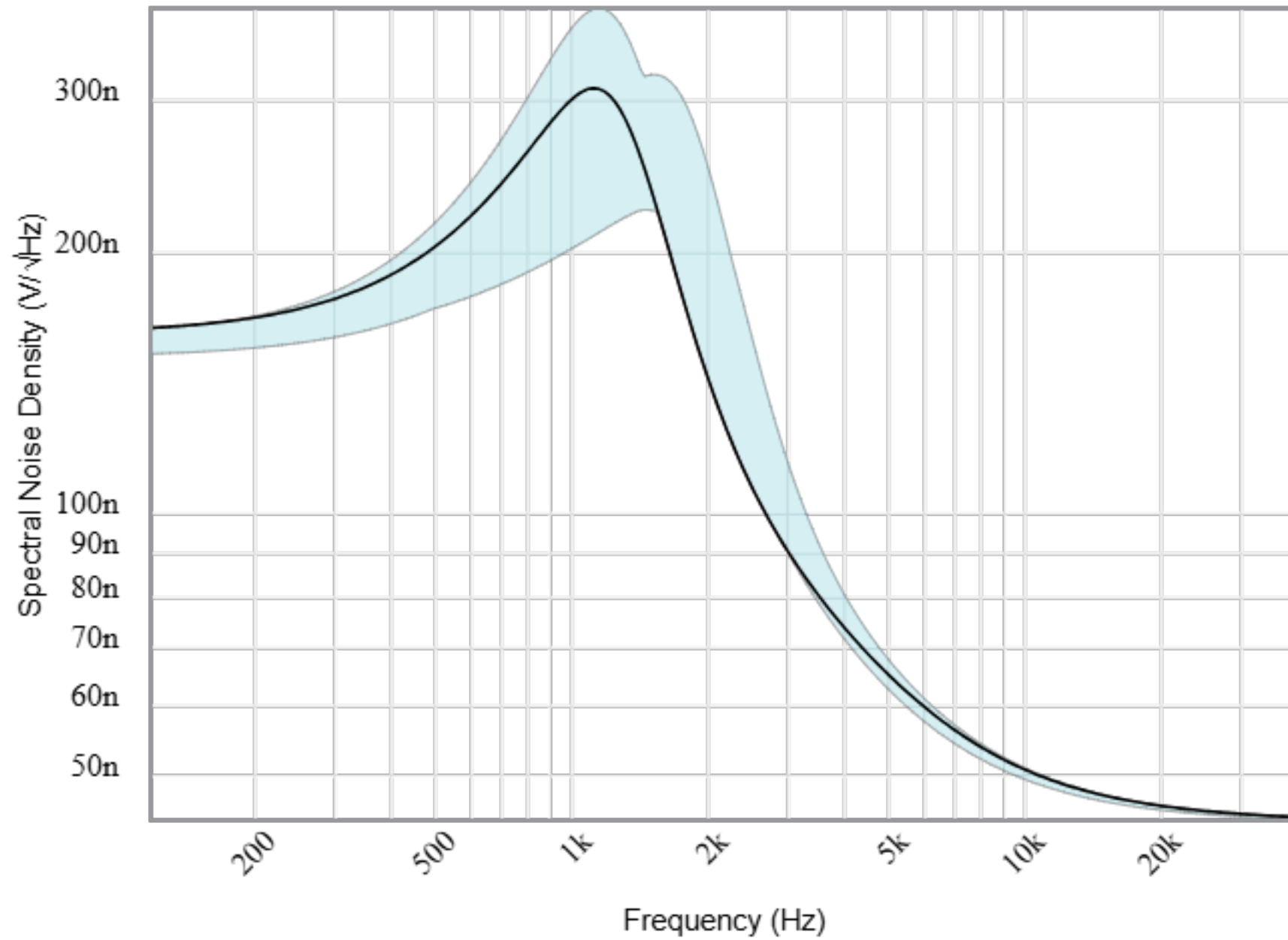
Input Impedance Magnitude



Input Impedance Phase



Noise



Stages

Your filter requires 2 op amp stage(s) with the following characteristics



2nd order
Low-Pass
Multiple
Feedback

Target

Simulated

Gain (V/V):

1

0.98 to 1.02

f_p (Hz):

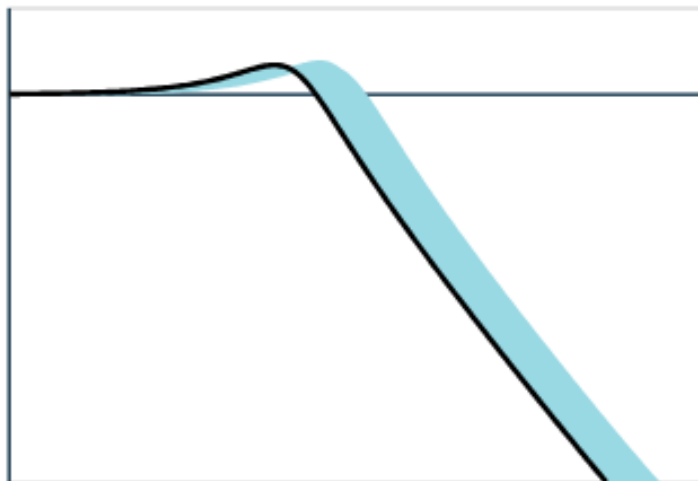
1.56k

1.53k to 1.92k

Q:

1.31

1.32 to 1.46



2nd order
Low-Pass
Multiple
Feedback

Target

Simulated

1

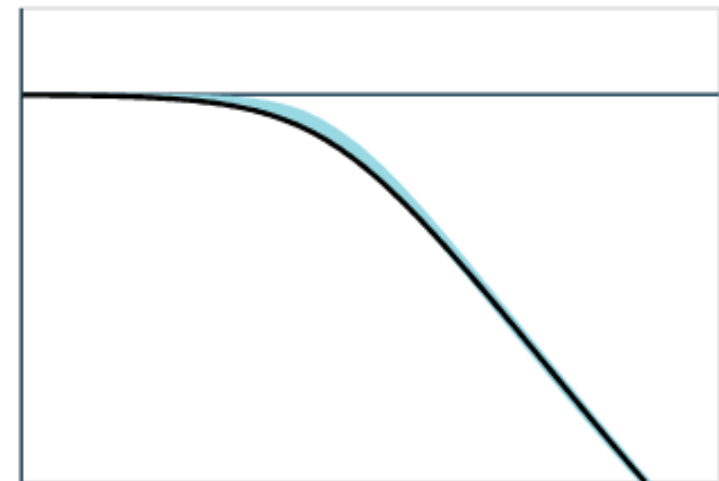
0.98 to 1.02

1.56k

1.45k to 1.64k

541m

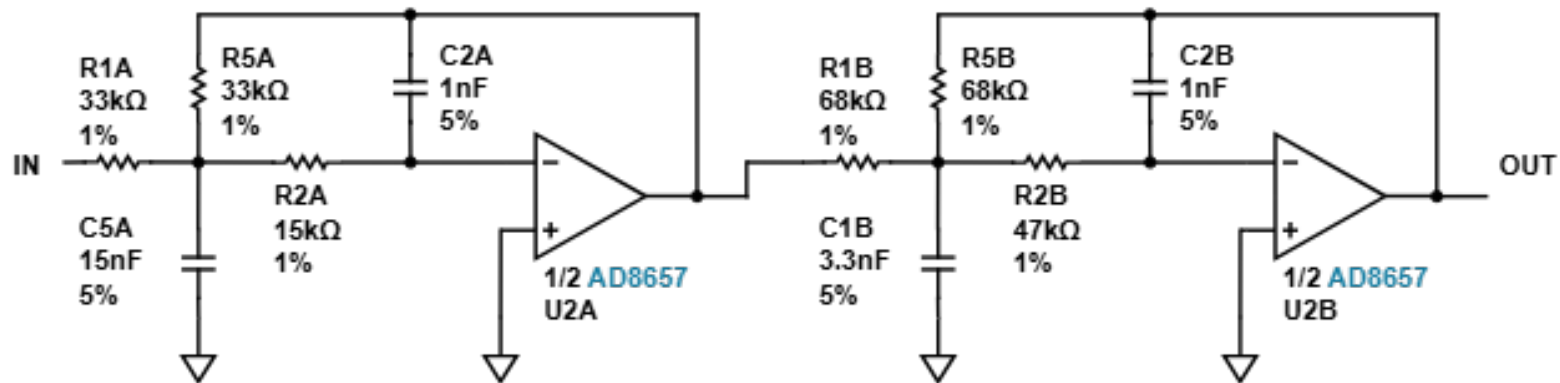
543m to 666m



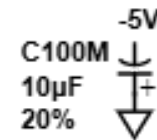
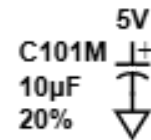
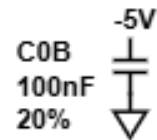
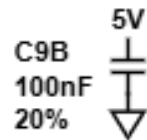
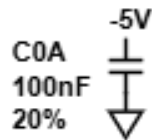
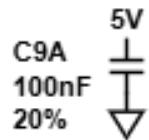
Circuit

Stage A
2nd order
Low-Pass
Multiple Feedback

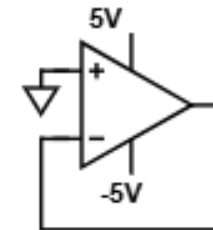
Stage B
2nd order
Low-Pass
Multiple Feedback



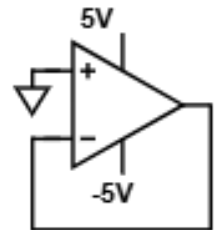
BYPASS CAPACITORS



SPARES Why The Spares?



1/2 AD8657
U2A



1/2 AD8657
U2B